Dual RSSI Indoor Localization In 3D Space

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***Abstract*—Bluetooth Low Energy (BLE) indoor localization appears to be a promising option to provide cost-effective solutions across various domains, including warehousing, healthcare, and security. This study aims to develop a 3D indoor localization system utilizing BLE technology in a static environment. The proposed method for device localization is based on fingerprinting, which involves utilizing dual RSSI values. Machine learning (ML) algorithms, such as K-Nearest Neighbor (KNN) and Support Vector Machine (SVM), were implemented and compared for their effectiveness. The impact on accuracy, by reducing the number of features was also examined using a Genetic Algorithm (GA). Additionally, the study aims to minimize the required physical measurements by employing interpolation techniques such as trilinear interpolation and nearest neighbor interpolation. In the experiment, a total of 196 reference fingerprints were collected. These reference points were then interpolated to a total number of 3610 fingerprints, resulting in three distinct data sets. These data sets were subsequently used to train the ML algorithms. Interpolation combined with KNN proved to be an efficient method, where the nearest neighbor interpolation reduced the average error for KNN from 134.8 cm down to 43.5 cm in a 4.5 m x 4.5 x m x 2.25 m room. The reduction of features by the GA on the other hand, proved to have little impact as it slightly worsened the system without being able to fully remove a beacon. Since the data set utilizes dual RSSI values, removing a feature will not automatically allow for the removal of a beacon, both paired features need to be deselected by the GA.**

***Index Terms*—3D Indoor Localization, Dual RSSI, BLE, Fingerprinting, KNN, SVM, Trilinear Interpolation, Nearest Neighbor Interpolation**

I. Introduction

Global Positioning System (GPS) is the universal standard regarding general localization. However, GPS is limited in regard to indoor localization since it relies on satellite signals where the penetration of walls and other similar obstructions is weak. Furthermore, GPS does not accommodate for height, which is essential for indoor localization systems in 3D. Therefore, other technologies have been developed and become predominant in the field of indoor localization.

Indoor localization Systems (ILS) have been extensively researched in recent years, due to their various applications in different environments and domains such as warehousing, healthcare, and security [1],[2]. Bluetooth Low Energy (BLE) beacons are popular, due to their low power consumption, low cost, and accessibility to support various devices. These characteristics make BLE an ideal choice for indoor localization research. Apart from BLE, other technologies such as Wi-Fi, ZigBee, Ultra-wideband (UWB), and Radio Frequency Identification (RFID) are also capable of enabling indoor location tracking of individuals or objects that are equipped with compatible devices.

This study introduces an approach to implement an indoor localization system in 3D space by integrating Dual Received Signal Strength Indicator (RSSI) with BLE. The main advantage of this approach is that it is cost-efficient and straightforward to develop applications that operate in the unlicensed industrial, scientific, and medical (ISM) radio bands. The main drawback is that a 2.4 GHz radio wave does not penetrate objects well, which makes measured signal strength unstable and jittery due to signal reflection, refraction, multi-path fading, etc [3].

The system employs the fingerprinting technique that maps the signal strength variations in a given area to predetermined locations. This allows for an accurate estimation of the locator device's true location. Since the system is static, line-of-sight issues can be disregarded, which further emphasizes the relevance of utilizing fingerprinting.

The vacant room designated as the experimentation area has the dimensions 4.5 m x 4.5 m x 2.25 m, and will be utilized for creating a fingerprinting radio map, collecting data samples, and implementing AI techniques. 196 out of 3610 fingerprints were collected physically. The remaining fingerprints were acquired by applying trilinear interpolation and nearest neighbor interpolation respectively. These two interpolation methods were then compared and evaluated.

The gathering of RSSI values was done with the Nordic Semiconductor SDK platforms, specifically the nRF52840 development kit (DK)[1] and nRF52840 dongles[2]. The wireless communication technology used was BLE, which is suitable for battery-powered devices due to its low power consumption.

This study aims to provide a thorough explanation of the implementation of a dual RSSI localization algorithm for BLE networks in a three-dimensional indoor environment. A discussion of the evaluation of the system and the metrics used to assess accuracy are included. Ultimately, this study could provide valuable insight for researchers and practitioners in further research and development to create a system that will work in a larger and more dynamic environment.

[1]https://www.nordicsemi.com/Products/Development-hardware/nrf52840-dk

[2]https://www.nordicsemi.com/Products/Development-hardware/nrf52840-dongle

II. Background

Wireless localization methods can be classified into two categories: signal strength-based probability and triangulation. The former method combines signal strength measurements to calculate the probability of the device's position, while the latter method calculates the position based on the distances between specific reference points [4]. In this study, the Received Signal Strength Indicator is a key parameter used to create the ILS.

There are two approaches to ILS using BLE beacons: range-based and range-free methods. The former method measures the distance between a receiver and multiple beacons, using techniques such as trilateration or multilateration [5]. The latter method does not require distance measurements and instead uses proximity-based techniques such as fingerprinting [6], [7]. Common range-based techniques based on Received Signal Strength Indicator, Angle of Arrival (AoA), Time of Arrival (ToA), and Time Difference of Arrival (TDoA) [8]. These techniques are used in both two-dimensional and three-dimensional ILS. The scope of this paper is on range-free localization where fingerprinting, which relies on signal strength of received signals rather than distance measurements, will be used.

A. Fingerprinting

Fingerprinting is a range-free technique used in ILS that utilizes wireless access technology such as BLE, Wi-Fi, or ZigBee [9]. It involves two distinct phases: offline testing and online training [10]. In the offline training phase, beacons gather RSSI data from predetermined positions, constructing a radio map that consists of fingerprints, representing each location where RSSI data is collected. During the online testing phase, the locator device measures real-time RSSI values received from nearby beacons. These RSSI values are then compared to the RSSI values stored in the radio map generated during the offline phase [11]. However, for the purpose of this project, the online testing phase will not be utilized. Instead, a machine learning (ML) model was trained to predict the location of the locator device based on the constructed fingerprint radio map generated during the offline phase.

B. Multilateration

Trilateration and multilateration are range-based techniques used in ILS to localize a target node by measuring its distances to at least three beacons with predetermined positions [12]. The process involves determining the intersection of three or more spheres that are centered at the devices and have radii equal to the measured distances. The intersecting points between these devices are considered the location of the target node.

C. Interpolation

Measuring a lot of coordinates in fingerprinting might not always be feasible. Interpolation techniques are one way of estimating values that lie between already measured values. Nearest neighbor interpolation assigns values of the nearest data point to be interpolated [13]. It selects the value from pre-existing data points that is the closest to the desired location. Trilinear interpolation is a three-dimensional interpolation method that estimates values within a volumetric dataset by considering the weighted average of the eight nearest neighboring data points [14]. This approach takes into account the values from the neighboring voxels along the respective axes to compute a smooth estimate for the desired point.

D. Machine Learning Algorithms for Indoor Localization

A variety of ML algorithms and techniques have been proposed for ILS. These approaches encompass both supervised and unsupervised learning ML methods, such as KNN, SVM and Neural Networks (NN).

KNN is a supervised learning algorithm that can be used in either classification or regression problems [11]. This ML model predicts the value of new data points by comparing the similarity of attributes in their value with the training data and finds out K neighbors which have maximum relevance with the new data point.

SVM is a supervised learning algorithm used for classification and regression tasks [11]. The algorithm aims to maximize the margin between the decision boundary known as the hyper-plane and the data points in a high-dimensional space. SVM maps the data points into this space and then determines the location of the hyper-plane that separates them into different classes. The placement of the hyper-plane is chosen to maximize the distance between the decision boundary and the closest data points of each given class. SVM is capable of handling a large number of characteristics and attributes and is capable of handling non-linear data, which works well for indoor localization [15].

III. Related work

The work presented in this report is focused on BLE technology together with RSSI. However, it is worth mentioning that indoor localization technologies such as Wi-Fi, ZigBee, RFID, and UWB have been extensively investigated in addition to BLE. According to Zafari et al.'s survey [8], UWB is considered the most accurate technology due to its high tolerance to interference. However, UWB requires additional hardware that can be costly. BLE, ZigBee, and RFID on the other hand, have advantages such as low power consumption, enabling the possibility of mobile node localization. Standards such as Wi-Fi and UWB require a higher power consumption and may not be suitable in the same application areas. One of the major challenges with BLE, ZigBee, and Wi-Fi is that they operate in the 2.4 GHz ISM band, which can introduce significant uncontrolled interference. Wi-Fi also operates on 5 GHz, and more recently also the 6 GHz band[3].

Lots of research has been done on range-based localization in BLE environments using RSSI values. The most common range-based methods are based on trilateration or fingerprinting. These methods allow for cheaper solutions than methods using AoA or ToA since they do not require additional

[3]https://ieeexplore.ieee.org/document/9442429

hardware to implement. Paterna et al. [16] utilized RSSI values for weighted trilateration in conjunction with Kalman filtering in an indoor setting, measuring both moving and static devices. When applying the methods, their precision was improved by $\sim$40% in different scenarios compared to basic trilateration. The precision for the static environment was up to 0.7 m in a 52 m^2 room.

Creating a radio map by taking densely located RSSI fingerprints to use as a dataset lends itself well for incorporation with supervised and unsupervised ML. Xiao et al. [17] applied the fingerprinting method together with a deep learning approach. In their work, they coupled their dataset with a deep-learning NN for regression and SVM for classification. The NN performed better and had on average less errors than the SVM. The results presented come from four different floors in a building where the floors were approximately 100 m x 100 m. For their NN algorithm, the mean error for the different floors was measured between 7 m to 8 m.

Compared to 2D localization environments, there has been relatively limited research conducted on 3D localization in indoor settings. Yang et al. [18] proposed a 3D localization scheme based on fingerprinting and a 1D Convolutional Neural Network (CNN). Similar to 2D approaches where dividing up a surface into 2D grids is the norm, they divided the space into a cubic grid. When utilizing a dense grid of fingerprints, this approach can be time-consuming. Furthermore, it can require thousands of fingerprint positions depending on room size and cube density. CNN is usually used in image classification problems, however, researchers have also tried to apply it to temporal signals. Yang et al. used the 1D CNN together with RSS time-series data from eight BLE beacons and their results showed a mean error of 25 cm in a 4 m x 2 m x 3 m room.

Cannizzaro et al. [19] presented a thorough study and comparison between the trilateration and fingerprinting methods. The methods were combined with ML techniques such as KNN, SVM, and multilayer perceptron (MLP). These ML methods were used to improve the accuracy of location tracking. Finally, the two different methods coupled with the ML techniques were evaluated in an industrial setting to test their effectiveness in a practical application. In their study, they presented results on the effect of beacon density, fingerprint density, and changes in the environment. Their results showed that in their setting, fingerprinting achieved a significantly higher precision compared to trilateration. In general, four beacons over three beacons resulted in minimal improvements. Four beacons gave a slight shift towards lower errors, however median errors became worse in some cases. Furthermore, creating a denser fingerprint grid did not linearly result in better precision. In their 8 m x 9 m environment, the precision clearly increased when reducing the spacing from 2 m to 1 m, but the gain was not as clear when reducing the spacing from 1 m to 0.5 m. The ML algorithms used performed with similar accuracy, therefore the authors argue that KNN outperforms the other algorithms due to its ease of use compared to SVM and MLP.

According to Nessa et al.'s survey [11] the most challenging problem regarding RSSI in ILS is the fluctuation in data leading to inadequate accuracy. Implementing ML proves to be useful in order to learn adequate information from the input data. For instance, Recurrent Neural Networks (RNN) could employ sequential correlation of time-varying RSSI measurements and use trajectory information to mitigate RSSI fluctuations.

Fingerprinting could be solved as a supervised learning problem by collecting and labeling individual fingerprints. However, it can also be solved with unsupervised approaches such as the ones in [20], [21], and [22]. By using unsupervised learning methods the work of labelling the data can be reduced. Specifically in [22] the fingerprints were crowdsourced, further reducing the exhaustive work of fingerprint collection.

IV. Problem formulation

The problem of indoor localization has been extensively researched, but existing technologies often require costly hardware and additional complexity. By utilizing RSSI values in BLE this study aims to close the gap in providing a cheap and straightforward alternative to 3D localization in static indoor environments. The main goal of this research paper is to investigate the accuracy of a three-dimensional indoor localization system. The system will utilize and combine fingerprinting of dual RSSI values in a 3D space together with ML algorithms such as KNN and SVM. The research questions this study aims to answer:

RQ1: What accuracy can be achieved in a controlled 3D environment using bi-directional fingerprinting RSSI values coupled with different ML algorithms?

RQ2: To what extent can a genetic algorithm maintain accuracy when reducing the number of beacons used for indoor localization?

Furthermore, since fingerprinting is a time-consuming process it is not feasible to measure thousands of points in a small space. This study aims to optimize the fingerprinting process by measuring fewer points and comparing different interpolation techniques. This aim leads us to the last research question:

RQ3: What is the effect on accuracy when using interpolation techniques for indoor 3D localization?

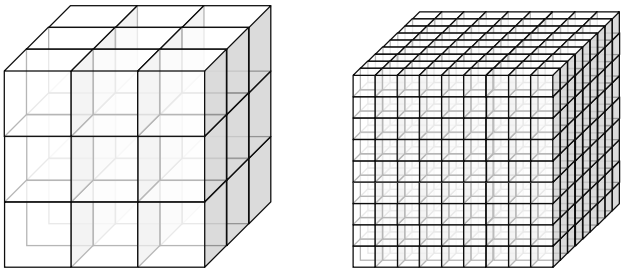
V. Method

The experiment was conducted in a 4.5 m x 4.5 m x 2.25 m empty room to avoid any line-of-sight interference. To create a data set, a fingerprinting scheme was constructed where the room was subdivided into a 25 cm x 25 cm x 25 cm cubic grid. However, the total number of vertices in the cubic grid was 3610, which would be impractical to measure individually if each fingerprint takes one minute, resulting in a total time of approximately 60 hours. To address this issue, an optimized scheme was implemented utilizing interpolated data. Figure 1 (a) shows a visual representation of the fingerprinting scheme, where the vertices are the points that will be measured. Figure 1 (b) shows the interpolated scheme where scheme (a) will serve as a base for the interpolation. In this case, 196 fingerprints were obtained physically and the remaining

fingerprints were acquired by employing trilinear interpolation and nearest neighbor interpolation respectively. Due to the nature of the localization system being volumetric, trilinear interpolation was a natural method to utilize. Nearest neighbor interpolation was implemented to compare the results. Three distinct data sets were compared:

1) A low density cubic grid with no interpolation.
2) Trilinear interpolation on the original data set.
3) Nearest neighbor interpolation on the original data set.

This method will answer **RQ3**, whether precise localization can be achieved with fewer measurements, thereby rendering the data collecting more time-efficient. The fingerprints obtained RSSI values from the bi-directional communication between the stationary beacons and the locator device. To accommodate for height, the locator device was positioned on an adjustable tripod stand.



(a) Base grid (b) Interpolated grid

Figure 1. Data set collection and interpolation of the cubic grid

The data set was collected with the objective of training and comparing two ML algorithms, namely KNN and SVM. The data set was labeled with positions and approached as a supervised learning problem. However, since the locations for new, unknown problems may not necessarily be discrete points such as those in the data set, the algorithms were chosen to be regression-based.

To answer **RQ1** the evaluation of accuracy was performed by comparing the real coordinates of the localization object and the predicted coordinates derived from the ML algorithms. The Euclidean distance formula seen in Equation 1), was applied to find the error in 3D space, where the distance between point p and q was measured, and n represents the number of dimensions.

$$d\left(p,q\right)=\sqrt{\sum_{i=1}^{n}\left(q_i-p_i\right)^2} \tag{1}$$

To answer **RQ2** a Genetic Algorithm was used to identify high-performing subsets of features for the different data sets. The purpose of this search was to investigate whether eight beacons are redundant for 3D localization, reducing the number of beacons implies the solution can be achieved with less cost and complexity.

VI. Experimentation

The experiment consisted of two parts: the data collection phase and the AI implementation phase.

A. Data Collection

The project utilized one nRF52840 development kit that was used as a central device and locator device. Eight nRF52840 dongles were used as beacon devices. Both of these devices can function as transmitters and receivers, enabling the collection of dual RSSI BLE values. The dongles were equipped with an integrated flash memory, which facilitated programming via USB connection. The software tools utilized in the project included Visual Studio Code, nRF Connect Programmer coupled with the Zephyr Bluetooth API.

The 25cm x 25cm grid was set up and marked with string on the floor that was secured by tape as shown in Figure 2 (a). The dongles were placed flat on platforms in the eight corners of the room and angled towards the center of the room to receive favorable measurements as seen in Figure 2 (b). The dongles were stationary and powered by a USB-A socket-to-plug adapter cable, which was connected to a power source. The mounts were printed for the dongles using Prusa i3 MK3 with PLA filament material which can be penetrated by radio waves[3]. The design was created in SolidWorks and exported as a .sli-file to PrusaSlicer ver 2.5.2 where the G-code was generated to create these mounts. Double-sided tape was used as an adhesive to fix the mounts onto the wall for longer periods. To secure the stationary placement of the locator device, another 3D-printed mount was created to sit on top of a tripod stand.



(a) Taped grid (b) 3D presentation of set up

Figure 2. Fingerprint setup

To create a fingerprint data set that stores the bi-directional RSSI values from the various positions of the locator device and the stationary beacons, a communication protocol between these devices was established. One nRF52840 DK was initialized as a BLE central device. This development kit was connected to a computer via UART through USB, and stationed on a 3D-printed platform on top of the adjustable tripod stand. The central and peripheral devices act as beacons which are periodically advertising in ten ms intervals while constantly scanning for advertisements. The devices filter advertisements based on specified MAC addresses. When the peripheral devices read a new advertisement from the central device, they embed the measured RSSI value into their next advertisement packet along with an identification number. In turn, the central device prints the dual RSSI values to the serial terminal in PuTTy[4] when it receives new advertisements since

[4]https://www.putty.org/

both values are embedded in the packets from the peripheral devices. The serial output was appended to a log file which was further used for preprocessing. The preprocessing consists of filtering out unwanted rows, filtering and averaging data, and also outputting a dataset to a .csv file.

Prior to conducting the measurements, it was necessary to determine orientation of the antenna for the locator device in order to receive favorable values during the data collection. A set of predetermined coordinates was used to collect a total of 24 measurements for each orientation. To uphold a consistent data collection process, the selected coordinates remained unchanged for the respective orientations. The orientations selected for comparison were along the X-axis, along the Y-axis, and at a 45-degree diagonal orientation. Following the data collection, heat maps were generated for each orientation, as depicted in Figure 3, in order to visualize the measurement strengths. The heat map corresponding to the Y-axis exhibited the most promising results, and was thereby chosen as the standard orientation for the data collection.

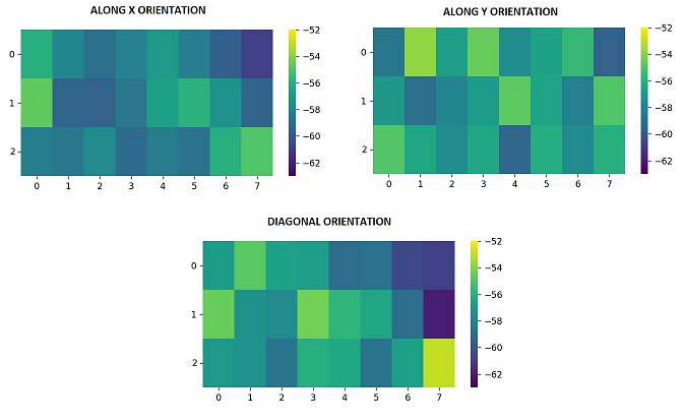


Figure 3. Heatmaps for different orientations.

A total of 196 reference fingerprints were collected physically and the remaining 3,414 fingerprints were interpolated using two methods: nearest neighbor interpolation and trilinear interpolation. For the nearest neighbor interpolation, a script searched for the four closest known fingerprints, determined their magnitudes, and normalized them to calculate their weights. These weights indicated the proximity of the four closest neighbors. Subsequently, the remaining values were interpolated and saved in an output file. A separate script was implemented for the trilinear interpolation. It worked by determining the RSSI value of a target point (x, y, z) by utilizing coefficients (c0 to c7) derived from the closest eight surrounding known points. The interpolation process involved applying a formula, where dx, dy, and dz represent the relative distances to the target point shown in Equation 2.

$$\begin{aligned}\text{Interpolated RSSI} = {} & c0 + c1 \cdot \text{dx} + c2 \cdot \text{dy} + c3 \cdot \text{dz} \\ & + c4 \cdot \text{dx} \cdot \text{dy} + c5 \cdot \text{dy} \cdot \text{dz} \\ & + c6 \cdot \text{dx} \cdot \text{dz} + c7 \cdot \text{dx} \cdot \text{dy} \cdot \text{dz}\end{aligned} \tag{2}$$

B. Machine Learning

The collected data set was first filtered; for every fingerprint location, the ten strongest measurements were kept and averaged. The reasoning was that the strongest RSSI values were the ones that were collected during the best conditions. The data set of the sparsely fingerprinted grid was interpolated using trilinear interpolation and nearest neighbor interpolation, creating three data sets in total. The different data sets were subjected to a GA. Since the ML algorithms were going to predict several labels, the GA ran on every axis and also employed both KNN and SVM as estimators to find feature subsets befitting both models. The population size was set to 50, approximately three times the size of the feature space. The algorithm ran for 50 generations, and it was designed to stop when no improvement was observed in the last five generations. The crossover probability was set to 95% while the mutation probability was set to 1%. Features were extracted for the X, Y, and Z-axis for both models and validated with 5-fold cross-validation.

The KNN model was trained using inverse distance weighting, where the Euclidean distance was employed for measurement. After tuning, the number of neighbors was set to three. Meanwhile, the SVM model utilized a radial basis function kernel, and the regularization parameter was tuned to ten, as it consistently demonstrated sufficient performance across all datasets. Both models were trained on each axis separately by removing the irrelevant labels for that specific axis, such as removing the "Y" and "Z" labels when training the X-axis. The models were also cross-trained using both interpolated and non-interpolated datasets. Additionally, the models were trained and compared after removing the features selected by the GA. Finally, during the prediction phase for the ML models, the result was validated by using 5-fold cross-validation.

VII. Results

By conducting the experiment, the following results were obtained in centimeters (cm), which are presented in Table I and Table II. The non-interpolated data set exhibits the highest average errors when compared to the two interpolated data sets. Without feature reduction, KNN and SVM exhibit similar average errors except for the nearest neighbor interpolated data set where KNN is twice as accurate as SVM. In the data sets where feature reduction was applied, the results are similar between KNN and SVM, with the exception that SVM shows a higher accuracy than KNN without interpolation.

By reducing the number of features, the results in Table I and Table II show that in general, higher accuracy is achieved without feature reduction except for the average error for SVM with no interpolation. No discernible patterns can be seen by performing feature reduction on the different models. In general, the GA removed different features for different models. It removed three to four features out of sixteen on average.

Table I
ACCURACY OF ML MODELS WITH NO FEATURE REDUCTION

	No interpolation	Trilinear interpolation	Nearest Neighbor interpolation
KNN min err (cm)	39.7	3.9	2.8
KNN max err (cm)	324.4	404.6	395.7
KNN avg err (cm)	134.8	57.7	43.5
KNN X avg err (cm)	64.3	29.4	21.7
KNN Y avg err (cm)	109.1	42.6	31.7
KNN Z avg err (cm)	44.7	25.5	20.5
SVM min err (cm)	39.9	5.4	6.3
SVM max err (cm)	338.0	361.2	350.5
SVM avg err (cm)	134.0	59.0	89.4
SVM X avg err (cm)	68.3	28.7	35.9
SVM Y avg err (cm)	104.8	45.4	71.8
SVM Z avg err (cm)	47.1	24.4	39.5

Table II
ACCURACY OF ML MODELS WITH REDUCED FEATURES

	No interpolation	Trilinear interpolation	Nearest Neighbor interpolation
KNN min err (cm)	20.7	6.1	2.3
KNN max err (cm)	374.3	413.3	426.3
KNN avg err (cm)	137.3	61.2	49.7
KNN X avg err (cm)	53.7	31.2	24.3
KNN Y avg err (cm)	105.9	42.9	32.8
KNN Z avg err (cm)	45.1	30.4	28.2
SVM min err (cm)	32.3	4.6	9.5
SVM max err (cm)	320.4	357.4	348.9
SVM avg err (cm)	118.9	66.9	93.9
SVM X avg err (cm)	56.7	32.1	37.7
SVM Y avg err (cm)	93.9	50.3	73.3
SVM Z avg err (cm)	44.8	30.2	47.6

VIII. DISCUSSION AND CONCLUSIONS

Based on the results presented in Tables I & II in Section VII, the accuracy achieved in the controlled 3D environment varies depending on the ML algorithm and interpolation method used. To answer **RQ1**, the combination of KNN, no feature reduction, and nearest neighbor interpolation yields the lowest average error with a minimum error of 2.8 cm, a maximum error of 3.97 m, and an average error of 43.5 cm. These results indicate the viability of employing ML in conjunction with bi-directional RSSI in controlled 3D environments.

Addressing **RQ2**, the results demonstrates that the impact of feature reduction can be deemed insignificant, and potentially inferior when compared to the inclusion of all features. Therefore, it can be concluded that feature reduction was unnecessary in this project for cost-saving and implementation complexity purposes. It can be argued that reducing features individually, as showcased in this report, adds complexity to the solution. Since dual RSSI measurements are utilized, the GA rarely removed a central and peripheral pair of RSSI values from the data sets. Removing a physical beacon would likely result in also removing a feature that is preferred by the GA.

To answer **RQ3**: The interpolated data results in lower errors than non-interpolated data, indicating that the interpolated RSSI data works well together with ML and holds potential for use in practical implementations in real-world applications. In particular, the nearest neighbor interpolation yields better results in terms of accuracy compared to trilinear interpolation, particularly when combined with KNN. However, SVM performs better when utilized together with trilinear interpolation than nearest neighbor interpolation. Nevertheless, it must be noted that these results are currently not validated using real RSSI values except for the 196 original measurements. This lack of validation poses a potential threat to the validity of the findings. On the other hand, increasing the density of fingerprints is widely recognized as a straightforward method to improve the accuracy of RSSI ML models. Therefore, interpolation can potentially be an efficient approach to further enhance the accuracy, even when a dense fingerprint map has already been collected physically.

Further improvements in results could be achieved by considering factors such as room size, fingerprint density, and alternative ML algorithms. Currently, the solution demonstrates promising scalability in terms of room size through interpolation and efficient data communication. The data communication relies on connectionless beacons that filter for noteworthy objects, facilitating a straightforward setup and integration of nodes. These nodes allow central units to enter or exit an area and provide RSSI values of peripheral devices.

IX. FUTURE WORK

The experiment was conducted in a static and controlled environment using fingerprinting. However, the future goal is to extend these ILS approaches and be able to apply them in the field where dynamic and uncontrolled environments are the norm. Further experiments have to be conducted in regard to optimizing data collection techniques to streamline the fingerprint collection process and reduce the effort required.

One way to expand the work could be to create detailed fingerprints for a smaller cube inside the cubic grid and extrapolate the relational data to the entire grid. This approach could be compared to creating an equally dense fingerprint map, allowing for a comparison of the results obtained from both methods.

In this project, the GA did not provide any improvement in regard to feature reduction. However, it is important to note that it cannot be confirmed or disproved whether having more or fewer beacons would result in better results. The chosen configuration of eight beacons, one in each corner, served as a baseline for conducting the experiment. While the configuration provided sufficient coverage, alternative configurations could yield better results. It would be beneficial to explore different geographical combinations and quantities of beacons.

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